STACKS, MODULI OF CURVES: LEMMA-SEMISTABLE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2373-2384 of sources/stacks-moduli-curves/pinned-original-source.tex; proof span is lines 2386-2472, the complete author proof (85 body lines).

Editorial note. Navigation stubs: 13 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 3 of queue_moduli-curves.json (pb=4061, stmt 2373-2384, proof 2386-2472).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0E3H: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2472–2472. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E3L: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2472–2472. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BS3: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2472–2472. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E63: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2472–2472. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BS6: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2472–2472. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E1J: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1128–1138. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:moduli-curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0E6N: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2043–2126. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E58: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2472–2472. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GG4: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2472–2472. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CTM: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2472–2472. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E6A: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2472–2472. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B40: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2472–2472. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E22: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2472–2472. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a prestable family of curves of genus $g \geq 1$. Let $s \in S$ be a point of the base scheme. Let $m \geq 2$. The following are equivalent*

1. *X_s does not have a rational tail (Algebraic Curves, Example 0E3H), and*
2. *$f^*f_*\omega_{X/S}^{\otimes m} \to \omega_{X/S}^{\otimes m}$, is surjective over $f^{-1}(U)$ for some $s \in U \subset S$ open.*

Proof. Assume (2). Using the material in Section 0E6N we conclude that $\omega_{X_s}^{\otimes m}$ is globally generated. However, if $C \subset X_s$ is a rational tail, then $\deg(\omega_{X_s}|_C) < 0$ by Algebraic Curves, Lemma 0E63 hence $H^0(C, \omega_{X_s}|_C) = 0$ by Varieties, Lemma 0B40 which contradicts the fact that it is globally generated. This proves (1).

Assume (1). First assume that $g \geq 2$. Assumption (1) implies $\omega_{X_s}^{\otimes m}$ is globally generated, see Algebraic Curves, Lemma 0E3L. Moreover, we have

$$\mathrm{Hom}_{\kappa(s)}(H^1(X_s, \omega_{X_s}^{\otimes m}), \kappa(s)) = H^0(X_s, \omega_{X_s}^{\otimes 1-m})$$

by duality, see Algebraic Curves, Lemma 0BS3. Since $\omega_{X_s}^{\otimes m}$ is globally generated we find that the restriction to each irreducible component has nonnegative degree. Hence the restriction of $\omega_{X_s}^{\otimes 1-m}$ to each irreducible component has nonpositive degree. Since $\deg(\omega_{X_s}^{\otimes 1-m}) = (1-m)(2g-2) < 0$ by Riemann-Roch (Algebraic Curves, Lemma 0BS6) we conclude that the H^0 is zero by Varieties, Lemma 0E22. By cohomology and base change we conclude that

$$E = Rf_*\omega_{X/S}^{\otimes m}$$

is a perfect complex whose formation commutes with arbitrary base change (Derived Categories of Spaces, Lemma 0CTM). The vanishing proved above tells us that $E \otimes^{\mathbf{L}} \kappa(s)$ is equal to $H^0(X_s, \omega_{X_s}^{\otimes m})$ placed in degree 0. After shrinking S we find $E = f_*\omega_{X/S}^{\otimes m}$ is a locally free $\mathcal{O}_S$-module placed in degree 0 (and its formation commutes with arbitrary base change as we've already said), see Derived Categories of Spaces, Lemma 0E6A. The map $f^*f_*\omega_{X/S}^{\otimes m} \to \omega_{X/S}^{\otimes m}$ is surjective after restricting to X_s. Thus it is surjective in an open neighbourhood of X_s. Since f is proper, this open neighbourhood contains $f^{-1}(U)$ for some open neighbourhood U of s in S.

Assume (1) and $g = 1$. By Algebraic Curves, Lemma 0E3L the assumption (1) means that ω_{X_s} is isomorphic to $\mathcal{O}_{X_s}$. If we can show that after shrinking S the invertible sheaf $\omega_{X/S}$ because trivial, then we are done. We may assume S is affine. After shrinking S further, we can write

$$Rf_*\mathcal{O}_X = (\mathcal{O}_S \xrightarrow{0} \mathcal{O}_S)$$

sitting in degrees 0 and 1 compatibly with further base change, see Lemma 0E1J. By duality this means that

$$Rf_*\omega_{X/S} = (\mathcal{O}_S \xrightarrow{0} \mathcal{O}_S)$$

sitting in degrees 0 and 1[1]. In particular we obtain an isomorphism $\mathcal{O}_S \to f_*\omega_{X/S}$ which is compatible with base change since formation of $Rf_*\omega_{X/S}$ is compatible with base change (see reference given above). By adjointness, we get a global section $\sigma \in \Gamma(X, \omega_{X/S})$. The restriction of this section to the fibre X_s is nonzero (a basis element in fact) and as ω_{X_s} is trivial on the fibres, this section is nonwhere zero on X_s. Thus it nowhere zero in an open neighbourhood of X_s. Since f is proper, this open neighbourhood contains $f^{-1}(U)$ for some open neighbourhood U of s in S. □

[1]Use that $Rf_*\omega_{X/S}^\bullet = Rf_*R\mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_X.\omega_{X/S}^\bullet) = R\mathcal{H}om_{\mathcal{O}_S}(Rf_*\mathcal{O}_X, \mathcal{O}_S)$ by Duality for Spaces, Lemma 0E58 and Remark 0GG4 and then that $\omega_{X/S}^\bullet = \omega_{X/S}[1]$ by our definitions in Section 0E6N.